

B. Sc. (Hons) Semester III Examination 2022-23
BOTANY
Paper No. BOB-301: Plant Ecology and Physiology

Time: 3 hours

Full Marks: 70

Note: Answer five questions in all, selecting two each from Sections A and B and Question 1 which is compulsory. Figures in the right-hand margin indicate marks.

1. **(Compulsory)** Answer the following questions in not more than 50 words each: [$10 \times 1 = 10$]
- (a) Synecology
 - (b) Net primary productivity
 - (c) Allogenic succession
 - (d) Food Web
 - (e) Ecological pyramid
 - (f) Guttation
 - (g) Phototropism
 - (h) What is apical dominance?
 - (i) Which products of light reaction of photosynthesis are used in the Calvin Cycle?
 - (j) H^+ -ATPase

Section - A

2. Write critical notes on any **three** of the following: [3 × 5 = 15]
- (a) Atmosphere
 - (b) Soil profile
 - (c) Qualitative Analytical Characters of a Community
 - (d) Biotic components of an ecosystem
3. Define population. Discuss in brief the various characteristics of a population of organisms. [15]
4. Write descriptive notes on any **two** of the following: [2 × 7½ = 15]
- (a) Ecotypes
 - (b) Phosphorus Cycle
 - (c) Universal model of energy flow
5. Define ecological succession. Discuss in detail the course of succession in a hydrosere. [15]

Section - B

6. Give an account of the C₄ and CAM pathways of CO₂ fixation in plants. [15]
7. Write descriptive notes on any **two** of the following: [2 × 7½ = 15]
- (a) Phytochromes
 - (b) Oxidative phosphorylation
 - (c) Photorespiration
8. Write short notes on any **three** of the following: [3 × 5 = 15]
- (a) Translocation of sugar
 - (b) Assimilatory reduction of nitrate
 - (c) Discovery of Auxins
 - (d) Biosynthesis of Gibberellins

External Portals & Academic Resources

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app/>